The cost of each distinction in the local computations of inter-universal Teichmüller theory IV

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Abstract

The local estimates of [IUT4, §1] are assembled from several steps that are kept apart in the source: a p-primary torsion term, a normalization that divides by the local degree, a tensor order sitting inside its own normalization, a floor function, a ramification support, and a weighting of places. Each separation is drawn explicitly there. This report computes, in actual local fields and for named curves, *what each of them is worth*: how much the estimate moves if that one distinction is elided and nothing else is changed. The answers are exact rationals or exact multiples of $\log p$ — $\log 3/2$, $\log 5/4$, $\log 5/4$ again for a different reason, $156/157$, a factor of 5.49×10^4, a weight of $2355/2356$ against $1/2$, and $\log 661$ at a single prime. `scripts/costs.py` computes twelve tables — standard library only, each asserting its own conclusion, exit zero — of which the ones displayed here are those the argument uses.

[S] marks a formula or hypothesis printed in the source and [P] an exact computation from it. *Nothing here is a correction to [IUT4].* Every distinction measured below is one the source itself makes — with the single exception of §8, noted there; the numbers say what an implementation loses by collapsing it, which is a question only an explicit datum can answer. *No claim is made about [IUT3, Cor. 3.12], about the objections of [SS], or about abc.*

1 What this is

[S] [IUT4, Thm. 1.10] fixes a collection of initial Θ-data as in [IUT1, Def. 3.1], but *it is not stated for an arbitrary one*: it supposes in addition that one is in the situation of [IUT3, Cor. 3.12], and that E_F has good reduction at every $v \in \mathbb{V}(F)^{\mathrm{good}} \cap \mathbb{V}(F)^{\mathrm{non}}$ not dividing 2ℓ. [P] Its proof passes through a chain of local statements — [IUT4, Prop. 1.1] through [IUT4, Prop. 1.7] — each of which separates quantities that a reader might be tempted to identify. This report evaluates them on named local fields and named curves. *Several of them are computable without a full initial Θ-datum* — and *some of the curves used below are not known to carry one*, the $N = 472$ curve of [Second, §13] being a curve with fourteen *candidate* levels. What the verified data of [First, Second] add is §10 and §12, where the parameters of an actual datum enter. [S] Modified versions of these local statements are derived in [Zhou, §1.1], and parts of them — the different in the unramified and tame cases, and the constants a_i, b_i of [IUT4, Prop. 1.2] — are formalized in Lean in Takkun-kohinata/IUT_LEAN [TK] (July 2026).

[P] The pattern is the same in every section. We take the source's own statement, instantiate it in a local field or at a curve that is named, compute the quantity exactly, then compute it again with one distinction removed, and print the difference. The difference is the cost of that

distinction. It is a number about an implementation, not about the theory: *a bound that keeps all the distinctions is correct whether or not the cost of any one of them is large.*

Remark 1 (why this is worth writing down)**.** [P] Two of the costs below were found because an earlier draft of [Second] got them wrong — an incorrect ramification index, and a formula applied where its hypothesis failed. In both cases what caught the error was the source's own statement being sharp enough to check against a specific curve. A table of costs is the form in which that sharpness becomes reusable.

2 Notation

[S] For a finite extension $k/\mathbb{Q}_p$ with ring of integers $\mathcal{O}$, ramification index e and residue degree f, we write $\mu^{\log}$ for the log-volume normalized by $[k:\mathbb{Q}_p]$, as in [IUT4, Prop. 1.4(i)] — the notation of [IUT4, Prop. 1.1, Prop. 1.2] supplies d_i and a_i, which enter later — so that $\mu^{\log}(\mathcal{O}) = 0$ and $\mu^{\log}$ of a sublattice of index p^m is $-\,m\log p/[k:\mathbb{Q}_p]$. All values below are given in units of $\log p$ unless stated otherwise.

3 The p-primary torsion term

[S] [IUT4, Prop. 1.4(ii)] gives, for a finite extension of $\mathbb{Q}_p$ with ramification index e, residue degree f, and p-primary torsion of order p^m in the units,

$$\mu^{\log}\big(\log \mathcal{O}^{\times}\big) = -\Big(\frac{1}{e} + \frac{m}{ef}\Big)\log p.$$

[P] Take $k = \mathbb{Q}_p(\zeta_p)$, so $e = p-1$, $f = 1$ and $m = 1$. Dropping the torsion term deletes $m/(ef)$:

p	e	f	m	$\mu^{\log}/\log p$	without m	cost
3	2	1	1	-1	$-1/2$	$1/2$
5	4	1	1	$-1/2$	$-1/4$	$1/4$
7	6	1	1	$-1/3$	$-1/6$	$1/6$

[P] So the cost is $\frac{1}{2}\log 3$, $\frac{1}{4}\log 5$, $\frac{1}{6}\log 7$: it is *not* uniform in p, and it is largest where p is smallest.

Remark 2 (correction)**.** [P] An earlier version derived these from the ceiling in a_i of [IUT4, Prop. 1.2]. The values were right and the derivation was not: a_i is a coefficient of the hull bound and m_i is the torsion variable, and they are different objects. The normalization itself is [IUT4, Prop. 1.4(i)].

4 The normalization by the local degree

[S] [IUT4, Prop. 1.4(i),(ii)] states its bounds for the normalized volume. [P] In the unramified extensions of $\mathbb{Q}_5$ of degrees $1, 2, 3$ the lattice index moves and the normalized volume does not:

f	index	raw log index$/\log 5$	$\mu^{\log}/\log 5$
1	5	1	-1
2	25	2	-1
3	125	3	-1

[P] Reading the raw index as a log-volume therefore counts the residue degree a second time, and the error grows linearly in f. This is the cheapest distinction to state and the easiest to lose, because both quantities are "the index".

5 The tensor order inside its normalization

[S] [IUT4, Prop. 1.1] and [IUT4, Prop. 1.4(iii)] distinguish an order from the normalization in which it sits. [P] Let $k = \mathbb{Q}_5(\sqrt{5})$, $\pi = \sqrt{5}$, and $R = \mathcal{O} \otimes_{\mathbb{Z}_5} \mathcal{O}$ inside $\mathcal{O} \times \mathcal{O}$. Then

$$R = \{(u, v) \in \mathcal{O}^2 : u \equiv v \bmod \pi\}, \qquad [\mathcal{O}^2 : R] = |\mathcal{O}/\pi| = 5, \qquad \dim_{\mathbb{Q}_5}(k \times k) = 4,$$

in which $\mathcal{O}^2$ is a $\mathbb{Z}_5$-lattice of rank 4 and $k \times k$ is the $\mathbb{Q}_5$-vector space it spans, so $\mu^{\log}(R) = -\frac{1}{4}\log 5$. [P] Treating R as its own normalization sets this to 0 and loses exactly $\log 5/4$. *Numerically this is the same as the cost of the torsion term at $p = 5$ in §3, namely $1/(p-1) = 1/4$, and it has a different origin;* the coincidence is a reason to keep the two apart rather than to merge them. *It is not the same as anything in §4*: there the $f = 4$ discrepancy between the raw and the normalized volume would be $3 \log 5$. [S] [DHPS, Rem. 6.2.1(2)] lists the index of $\bigotimes_i \mathcal{O}_{K_i}$ in the product $\mathcal{O}_L$ of the rings of integers of the field components as a quantity one could compute, and says that this "seems practical to do in specific toy cases"; the index 5 above is such a case.

6 The floor

[S] [IUT4, Prop. 1.4(iii)] passes from the container to the smooth bound by $-\lfloor A \rfloor \le -A + 1$, so the slack is $1 + \lfloor A \rfloor - A$. [P] In the moderate tame model here $d_I + a_I = |I|$ is an integer, so the slack is $1 + \lfloor \lambda \rfloor - \lambda$:

λ	$1 + \lfloor \lambda \rfloor - \lambda$	$\lceil \lambda \rceil - \lambda$ (wrong)
3	**1**	0
472/157	156/157	156/157
627/157	1/157	1/157
4	**1**	0

Remark 3 (correction)**.** [P] An earlier version used $\lceil \lambda \rceil - \lambda$. The two agree off the integers and differ *on* them: at $\lambda = 3$ and $\lambda = 4$ the slack is 1, not 0.

[P] At $\lambda = 472/157$ — the value carried by the curve of [Second, §13], which has *fourteen candidate levels* — the slack is 156/157, while the stage-one entry margin of that same curve is 1/157. *Their ratio is* 156*, and they are not comparable quantities at all:* one is a defect between a container and a smooth bound, the other a margin between the two sides of a comparison. Placing them side by side without saying which is which is easy to lose, and that is why the two are printed together here.

7 The ramification support of the input

[S] [IUT4, Thm. 1.10], Step (iii), and [IUT4, Rem. 1.10.5(iii)] work with the ramification support of the input. *Step (viii) then bounds that support from above by a sum over all primes below a cutoff*, so the source draws both, and what is measured here is the size of the gap between them at one

curve. An earlier version of this paragraph said the source used the support *rather than* all primes; that misdescribed Step (viii) (found in a later review, 2026-09-07). [P] Take the curve `5190c3`, with

$$\Delta = 2^6 \cdot 3^{44} \cdot 5 \cdot 173,$$

and the torsion field $K_0 = \mathbb{Q}(i, E[210])$, that is, $F(E[7])$ for the field $F = \mathbb{Q}(i, E[30])$ that [DH25, §§2.10–2.11] uses for elliptic curves over $\mathbb{Q}$. Its ramification support is $\{2, 3, 5, 7, 173\}$ — five primes. *That this list is exactly the support is an ordinary number-theoretic statement about the curve and the torsion field, not a consequence of the factorization.* It is proved as follows. Away from those five primes E has good reduction and 210 is invertible, so by the criterion of Néron–Ogg–Shafarevich the field K_0 is unramified there. Conversely 2 ramifies in $\mathbb{Q}(i)$; the Weil pairing puts $\mathbb{Q}(\mu_{210})$ inside K_0, and $3, 5, 7$ ramify in it; and at 173 one has $v(\Delta) = 1$ with c_4 a unit, so the reduction is multiplicative and the Tate parameter has valuation 1 after at most an unramified quadratic extension — rationalizing the 3-torsion then requires an element of valuation $1/3$, so 173 ramifies as well. Both inclusions follow. [P] What the script checks is the arithmetic this rests on: what the script checks is that it is consistent with Δ and with the ramification of $\mathbb{Q}(i, \mu_{210})$. The number of *all* primes below the cutoff 3 870 720 is 274 666, computed by sieve, with no appeal to any effective prime number theorem. The ratio is

$$\frac{274666}{5} = 54933.2.$$

[P] So replacing the input support by the all-prime proxy that Step (viii) uses multiplies the count of primes by 5.49×10^4 at this curve. *It is the only cost tabulated here that is a factor rather than an additive term*; the comparison is of counts of primes, not of every quantity in the report.

8 The weighting of places

[S] [IUT4, Rem. 1.7.1] weights the places of the moduli field by their local degrees over $[F_{\mathrm{mod}} : \mathbb{Q}]$. [P] *Weighting them by their ramification indices instead is not an alternative the source anywhere draws*: unlike the other sections here, this one measures an elision that only an implementation could make. At a split prime with $e_{\mathrm{b}} = 2355$ and $e_{\mathrm{g}} = 1$:

weight from e alone, bad side	2355/2356
correct weight, bad side	1/2
ratio	2355/1178
two-factor tuples gg, gb, bg, bb	1/4 each

[P] Weighting by e sends the bad side to 2355/2356, that is, to almost all of the mass, where the correct weight gives it half. The auxiliary degrees cancel; the place count does not.

9 A mean that agrees while the components do not

[S] [IUT4, Prop. 1.7] sums over ordered tuples, and [IUT4, Rem. 1.10.7(ii)] keeps the mixed components. [P] In $(\mathbb{Q}_p \times \mathbb{Q}_p) \otimes (\mathbb{Q}_p \times \mathbb{Q}_p)$ there are four components. *Assigning them the illustrative values* $0, 8, 8, 16$ — these are chosen to display the phenomenon, not read off a datum — the mean is 8, and the two diagonal components alone, with $0, 16$, have mean 8 as well.

[P] *The means agree.* What differs is the probability carried by a mixed component: $1/2$ against 0. So a check on the mean cannot detect the collapse of the mixed components, and any verification of an implementation that compares only averages will pass a program that has lost them.

10 Two boundaries in the explicit families

[P] $e_{\mathrm{b}} \le p-2$ is *not* a hypothesis of [IUT4, Prop. 1.4(iii)]: there I^* is a chosen subset and the inequality is asked only outside it. It is the condition under which the place may be left out of I^*, that is, the condition for the error-free form of the bound. [P] For $\pi = 41+26i$, $p = 2357$, $\ell = 157$ and $a = \pi^N$ one has $e_{\mathrm{b}} = 15\ell/\gcd(15\ell, 2N)$, so a single step in N moves it by a factor of 471:

N	$\gcd(2355, 2N)$	e_{b}	$e_{\mathrm{b}} \le 2355$
471	471	5	yes†
472	**1**	**2355**	yes
474	3	785	yes
475	5	471	yes

[P] †*The row $N = 471$ is a local arithmetic candidate only*: there $\ell = 157$ divides $2N = 942$, so [IUT1, Def. 3.1(c)] is not satisfied and it is not a datum to compare with. The row is printed because the gcd computation is the point. The curve of [Second, §13] — the one with *fourteen candidate levels*, not fourteen verified initial Θ-data — has $N = 472$, where the gcd is 1 and e_{b} is maximal.

[P] The second boundary concerns gcds of ideals against gcds of norms. For $\pi = 16+3\sqrt{5}$ and $N = 110$ the prime 661 splits, and $\sqrt{5}$ has the two roots 115 and 546 modulo 661. *Those residues are not square roots of* 5 *to higher precision* — $115^2 - 5 = 20 \cdot 661$ is not 0 modulo 661^2 — so they are Hensel-lifted before use; deciding that a valuation is exactly 1 needs a genuine local square root, not a root modulo 661. Reducing $a = (16+3\sqrt{5})^{110}$ and $t = 1-a+a^2$ at each lifted root:

$\sqrt{5} \bmod 661$	$a \bmod 661$	$v(1-a)$	$v(t)$
115	1	1	0
546	365	0	1

[P] *The* 661 *of the numerator and that of the denominator sit at different primes.* The total norms nonetheless carry 661 twice, so cancelling them removes it once too often. Writing $D = \mathrm{N}(a^2b^2)$ and $U = \mathrm{N}(256t^3)$ for $b = 1-a$, and taking the Legendre invariant $j = 256t^3/(a^2b^2)$, the two finite heights normalized by the degree are

$$H_{\mathrm{ideal}} = \tfrac{1}{2}\log\frac{D}{G_{\mathrm{ideal}}}, \qquad H_{\mathrm{norms}} = \tfrac{1}{2}\log\frac{D}{\gcd(U,D)},$$

where $G_{\mathrm{ideal}} = \mathrm{N}\big((256t^3, a^2b^2)\big) = 2^8$. *The identity $t+ab = 1$ fixes the support of the common factor, not its exponent*, so the 2-adic valuation has to be computed. Since t and ab share no prime ideal, the only common prime is the 2 coming from 256; and 2 is inert in $\mathbb{Q}(\sqrt{5})$, so $v_{(2)}(x) = \frac{1}{2}v_2(\mathrm{N}\,x)$. Here $a \equiv 5$ and $b \equiv 4 \pmod 8$, whence $v_{(2)}(a) = v_{(2)}(t) = 0$ and $v_{(2)}(b) = 2$. Therefore $v_{(2)}(256t^3) = 8$ and $v_{(2)}(a^2b^2) = 4$, so the ideal gcd is $(2)^4$ and its norm is 2^8. The norm gcd carries 661^2 in addition. Both are computed from the curve in `scripts/costs.py`, giving

$$H_{\mathrm{ideal}} = 1174.636200642495, \qquad H_{\mathrm{norms}} = 1168.142446802643, \qquad H_{\mathrm{ideal}} - H_{\mathrm{norms}} = \log 661$$

to twelve places. *The reason is in the table, not only in the decimals*: a check carried out on the norms alone cannot see which prime a valuation belongs to.

11 A finite statistic for the tame local coefficient

[P] *This section is about the coefficient of* $\log p$ *in the simplified bound, not about an actual possible-image log-volume.* Fix a stage cutoff $m \geq 2$ and write

$$T_m = \coprod_{j=1}^{m} \{\mathrm{g}, \mathrm{b}\}^{j+1}, \qquad c_{\mathrm{g}} = 1 - e_{\mathrm{g}}^{-1}, \quad c_{\mathrm{b}} = 1 - e_{\mathrm{b}}^{-1}, \quad s = \mathrm{ord}_p(\underline{\underline{q}}) > 0.$$

[P] In the tame specialization with every $e_i \leq p-2$ — so that $I^* = \varnothing$ is available, which is a *specialization* of [IUT4, Prop. 1.4(iii)] and not a hypothesis satisfied by arbitrary tuples — the coefficient collapses to two counts and one indicator:

$$\boxed{B(t) = 1 + c_{\mathrm{g}} n_{\mathrm{g}}(t) + c_{\mathrm{b}} n_{\mathrm{b}}(t) - s\, j(t)^2 \mathbf{1}_{\{t_j = \mathrm{b}\}}}$$

Proposition 1. [P] *Put* $\pi(t) = (n_{\mathrm{b}}(t), \perp)$ *when* $t_j = \mathrm{g}$ *and* $\pi(t) = (n_{\mathrm{b}}(t), j)$ *when* $t_j = \mathrm{b}$. *On* T_m *with* $m \geq 2$, B *factors through* π *if and only if* $e_{\mathrm{g}} = 1$. *If moreover* $e_{\mathrm{b}} > 1$ *and* $s > 2c_{\mathrm{b}}$, *the induced map is injective.*

Proof. Sufficiency is substitution. For necessity, (g, g) and $(\mathrm{g}, \mathrm{g}, \mathrm{g})$ have the same π and their B differ by c_{g}; this uses $m \geq 2$. For injectivity write $c = c_{\mathrm{b}} > 0$: the good-ending values are ≥ 1 while the largest bad-ending value at stage 1 is $1 + 2c - s < 1$, and the maximum at stage $j+1$ minus the minimum at stage j is $c(j+1) - s(2j+1) < 0$, so the stages occupy disjoint ranges. □

[P] *The domain is part of the statement.* At a *fixed* stage $n_{\mathrm{g}} = j + 1 - n_{\mathrm{b}}$, so B factors through π for every e_{g} and the factorization says nothing; the equivalence lives on T_m, where the two counts are independent. The strict inequality is also needed: at $s = 2c_{\mathrm{b}}$ the classes $(0, \perp)$ and $(2, 1)$ take the same value.

Remark 4 (the exact criterion, and why $s > 2c_{\mathrm{b}}$ suffices)**.** [P] The condition $s > 2c_{\mathrm{b}}$ is *sufficient but not necessary.* Assume $e_{\mathrm{g}} = 1$, $c_{\mathrm{b}} > 0$ and $s > 0$, and put $r = s/c_{\mathrm{b}}$ and

$$E_m = \left\{ \tfrac{d}{j^2} : 1 \leq j \leq m,\ 1 \leq d \leq j+1 \right\} \cup \left\{ \tfrac{d}{k^2 - j^2} : 1 \leq j < k \leq m,\ 1 \leq d \leq k \right\}.$$

Then *the induced map is injective if and only if* $r \notin E_m$: a good-ending class meets a bad-ending one at stage j exactly when $r = (n-h)/j^2$ with $1 \leq n - h \leq j+1$, and two bad-ending stages $j < k$ meet exactly when $r = (n_k - n_j)/(k^2 - j^2)$ with $1 \leq n_k - n_j \leq k$; within one group $c_{\mathrm{b}} > 0$ separates. *This criterion arose in an internal check of the present note.*

[P] It also explains the crude condition. The first family is maximised at $j = 1$, $d = 2$, giving 2; the second is bounded by $k/(k^2 - (k-1)^2) = k/(2k-1) < 1$. Hence $\sup E_m = 2$ *for every* m, so $r > 2$ — that is, $s > 2c_{\mathrm{b}}$ — already misses E_m at every cutoff. For the profile of [Second], $r = \frac{29}{7} / \frac{104}{105} = \frac{435}{104} > 2$.

[P] For the profile of [Second] — $e_{\mathrm{g}} = 1$, $e_{\mathrm{b}} = 105$, $s = 29/7$, so $c_{\mathrm{b}} = 104/105$ and $2c_{\mathrm{b}} < s$ — the hypotheses hold, and T_3 has 28 tuples with 13 distinct values. *So the good places are invisible to this coefficient*: it sees how many bad places there are, and, when the last place is bad, at which stage it is applied.

Remark 5. [P] This is a minimal statistic *for the single function* B: its fibres agree with those of π under the stated hypotheses. It is not a claim that the statistic supports any later operation in the source's proof, and it is not a statement about actual log-volumes.

12 Two items recovered

[P] Two items were listed as not established in an earlier version of this report. Both are now settled, and in each case the author's own computation had been correct while the quantity being compared was not the one reported.

The two-adic boundary. [P] For $\pi = 10 + 3i$ the finite height contribution at 2 is not v_2 itself but the remainder left by a cap at 8. *The cap is ours, and it comes from this family, not from a general statement*: [IUT4, Cor. 2.2(i)] defines $\log(q^{\forall})$ and $\log(q^{\neq 2})$ and prints no such cutoff. At the prime $\mathfrak{p} = (1+i)$ of $\mathbb{Q}(i)$ one has $v_{\mathfrak{p}}(2) = 2$ and $v_{\mathfrak{p}}(a) = v_{\mathfrak{p}}(1-a+a^2) = 0$, so from $j = 256t^3/(a^2b^2)$,

$$v_{\mathfrak{p}}(j) = 16 - 2v_{\mathfrak{p}}(1-a), \qquad v_{\mathfrak{p}}(1-a) = v_2\big(\mathrm{N}(1-a)\big),$$

and the normalized local finite height $\frac{1}{2}\max\{-v_{\mathfrak{p}}(j), 0\}\log 2$ is

$$\max\big\{v_2(\mathrm{N}(1-a)) - 8,\ 0\big\}\log 2.$$

N	$v_2(\mathrm{N}(1-a))$	$\max\{v_2-8,0\}$	contribution
199	1	0	0
200	9	1	$\log 2$
201	1	0	0
400	11	3	$3\log 2$

[P] *The values* $1, 9, 1, 11$ *computed earlier were right*; they are v_2, and the reported quantity is the excess over the cap. One step in N moves the contribution between 0 and $\log 2$.

Two permitted upper bounds at one prime. [P] At $\ell = 157$, $p = 2357$ one has $e_{\mathrm{b}} = 2355$ and $e_{\mathrm{g}} = 1$, both $\leq p - 2 = 2355$, so $I^* = \varnothing$ is available and the extra error term of [IUT4, Prop. 1.4(iii)] is *exactly* 0. The uniform coarse term, averaged over j, is

$$(\ell + 5)\log(e^*_{\mathrm{mod}}\ell) = 162\log(1105920 \cdot 157) = 3073.534296832696\ldots$$

[P] *This compares two permitted upper-bound expressions at one fixed prime*, averaged over j. It is not the actual indeterminacy, and it is not a gap in [IUT3, Cor. 3.12]; no such claim is made here.

13 What is not established here

[P] *Nothing remains on this list.* The two items an earlier version placed here — the two-adic boundary and the comparison of a precise against a coarser local error term — are established in §12. They are recorded here because the reason for the earlier omission is worth keeping — *both are resolved in §12 of the present note and neither remains open* — in both cases the author's arithmetic was correct and the *quantity being compared* was not the one reported, so the check failed for a reason that no amount of recomputation would have found.

14 Scope

Claimed. The values printed above, as exact rationals or exact multiples of $\log p$, for the local fields and curves named with them.

Not claimed. That any statement of [IUT1, IUT3, IUT4] is in error; that any estimate in those papers is affected; any conclusion about the *abc* inequality; that the objections of [SS] or [IUT3, Cor. 3.12] are touched by anything here. No log-volume of an actual Θ-pilot object is computed, no hull is compared, and no membership in any exceptional set is addressed. *The costs above are costs of eliding a distinction in an implementation; they are not defects of the source, which draws every one of those distinctions explicitly and is cited for each.*

Reproduction

`scripts/costs.py` prints all twelve tables using only the standard library, asserts each conclusion — including the sieve count $\pi(3\,870\,720) = 274666$, the factorization of Δ, the four values of e_{b}, and the agreement of the two means — and exits zero. *Table 12 checks every clause of Proposition 1*: the equivalence with $e_{\mathrm{g}} = 1$ on T_3, that at a *fixed* stage the factorization holds for every e_{g} and so would be vacuous there, the injectivity under $e_{\mathrm{b}} > 1$ and $s > 2c_{\mathrm{b}}$, the collision of $(0, \perp)$ with $(2, 1)$ at $s = 2c_{\mathrm{b}}$ exactly, and the counts $|T_3| = 28$ with 13 distinct values. *That proposition is the one statement here that was wrong twice before it was right* — first in its domain, then in its injectivity hypothesis — which is why each clause is asserted rather than only stated.

[P] *Four different kinds of warrant appear in this note and we do not run them together. (1)* Exact rational or integer computation: the tables, and the two heights of §10, computed from the curve. *(2)* Finite exhaustive check over a stated range: Table 12's clauses, verified on T_3 and on the rational r enumerated there — a check on a finite set, not a proof for all m. *(3)* Ordinary proof: Proposition 1 and Remark 4, which are proved for all m in the text and are *not* formalised anywhere. *(4)* Kernel verification, *narrowly*: in [Second] the files `ImaginaryQuadraticDatum.lean` and `ModSevenImage.lean` re-prove the integer identities and finite predicates of the datum $a = (10+3i)^{29}$ — norm factorisations, valuations, point counts and traces at the witness places, and the finite group computations — and `FamilyArithmetic.lean` the finite algebra behind that report's existence family. *No statement of the present note is among them*, and nothing here is discharged by them. In particular the curve of [Second, §13] with $N = 472$ carries *fourteen candidate levels*, not fourteen verified initial Θ-data. [First] has no kernel check of its datum: its statements for $a = (16 + 3\sqrt{5})^{29}$ are checked by the programs `scripts/checks.py`, `scripts/check_round3.py` and `scripts/check_normalization.py` and by the exhaustive sieve `scripts/check_norm_seventh_power_free.cpp` (`scripts/check_crt_family.py` checks the separate existence family of its companion draft), and its only Lean file, `lean/RegardedUpToIsOrbit.lean`, is a finite model that concerns no datum. Earlier versions of this note listed those statements under (4) as kernel-verified; they are not.

[P] Within (1), one more line is worth drawing: D, U, G_{ideal} and $\gcd(U, D)$ are exact integers and the displayed formulas for H_{ideal}, H_{norms} are exact; the twelve-place decimals printed for them are floating-point evaluations of those exact expressions.

Disclosure

Large language models were used substantially in locating passages, carrying out computations, drafting and adversarial review; the author is responsible for the claims. Every number above was

recomputed by the author independently of the process that first produced it, and the two items where an earlier recomputation had failed were recorded rather than dropped silently; both are settled in §12.

Version 1.0.3 corrects the description of [First] under (4) of the Reproduction section, cites [DH25] in §7 (earlier versions listed it without citing it), and cites earlier work: [Zhou] and [TK] in §1, and [DHPS] in §5. No value is changed.

References

[First] T. Urahigashi, *An explicit initial Θ-datum with mixed reduction at a split prime*, Zenodo, DOI 10.5281/zenodo.22537342, 2026.

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